

Explainable Legal Outcome Prediction for the ECtHR: A Two-Stage Argument Mining Pipeline

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Abstract

The report describes a two-stage explainable procedure for predicting which articles of the European Court of Human Rights (ECtHR) were breached in a particular case. In the first stage, LegalBERT is fine-tuned using 12,947 annotated (case, article) pairs in order to identify the legally relevant sentences (the premises), attaining a macro F_1 score of 0.96 on the argument classification task. In the second stage, the premises thus extracted are embedded using a SentenceTransformer and a linear SVM is trained with per-article threshold adjustment on the LexGLUE ecthr_a benchmark. The hybrid method — which combines the full text with signals that take account of the premises — achieves 0.5680 macro F_1 , matching the full-text baseline (0.5669), while also providing explanations at the sentence level. An assessment of faithfulness using ERASER metrics shows that the premises cited are 4.4 times more influential than random ones, and 86.3% of the predictions can be reconstructed from just three of the cited sentences. A supplementary hierarchical study of LegalBERT seed configurations (with 5 seeds) yields a macro F_1 score of 0.6342 ± 0.0204 , the hybrid and full-text models being statistically indistinguishable ($p = 0.74$).

1 Introduction

Each year the European Court of Human Rights (ECtHR) receives thousands of applications, all of which have to involve a decision as to which of the Convention’s articles have been breached. It would be possible to automate some of this process and thus improve access to justice by enabling applicants to evaluate the strength of their cases and assisting legal professionals in prioritising them. Nevertheless, any system introduced into a legal context has to be explainable: otherwise a prediction lacking a traceable rationale has only limited practical use and gives rise to serious

accountability issues.

Previous research has mainly treated the task of predicting ECtHR outcomes as one of text classification. Chalkidis et al. (Chalkidis et al., 2019) were the first to show that neural models could predict violations based on the facts of a case with reasonable accuracy. The LexGLUE benchmark (Chalkidis et al., 2022) then established a standard for evaluation on the ecthr_a task, with hierarchical Transformer models attaining state-of-the-art performance. Although these end-to-end models are powerful they lack transparency: they provide no clear way of determining which sentences led to a particular prediction. Attention weights have been suggested as a means of achieving explainability (Jain and Wallace, 2019), but it has been found that they are unreliable as indicators of the importance of features (Wiegrefe and Pinter, 2019).

We are putting forward a two-stage pipeline which builds explainability right from the start. In the first stage, a domain-specific LegalBERT (Chalkidis et al., 2020) is fine-tuned in order to classify each individual sentence as either a legally relevant premise or a non-premise, using 12,947 annotated ECtHR (case, article) pairs (azminajid, 2024) and including similarity-filtered factual negatives in order to prevent over-extraction. In the second stage, the premises that have been extracted are embedded using all-mpnet-base-v2 (Reimers and Gurevych, 2019) with concatenated pooling and a linear SVM (Pedregosa et al., 2011) is trained for each article. Since the SVM decision function is linear in the input features, each prediction can be attributed to specific sentences by projecting their embeddings onto the weight vector. Because the embedding includes non-linear operations such as max-pooling, this projection is an approximation rather than an exact decomposition, but it requires

no separate post-hoc surrogate model. We include the classifier as a supplementary deep-learning comparison.

We carry out experiments using the LexGLUE `ecthr_a` test set (1,000 cases, 11 classes: the full text (baseline), the premise-only case, and the hybrid case). In order to verify the premise extractor’s performance beyond that shown by held-out classification accuracy, we use matched-budget controls (DeYoung et al., 2020) which involve comparing premises with their complements, with random sentences, and with lead sentences all at the same word budget. We also assess the faithfulness of the ERASER comprehensiveness and sufficiency metrics (DeYoung et al., 2020). The BERT comparison adheres to the full LexGLUE seed protocol (5 seeds $\times$ 3 models).

The key results are that (i) the hybrid SVM achieves 0.568 macro F_1 , matching the full-text baseline (0.5669), while also providing sentence-level explanations; (ii) the premise attributions are accurate since premises that are cited have four point four times as much impact as random ones and 86.3% of the predictions involve just three cited sentences; (iii) hierarchical LegalBERT attains a macro F_1 of 0.634 ± 0.020 , the results for the hybrid and full-text models being statistically indistinguishable ($p = 0.74$); and (iv) the extractor is able to identify sentences that have an argument-like character but argument-likeness and the signal predictive of the article are not the same, indicating a fundamental challenge in the mining of legal arguments.

2 System description

2.1 Overall Pipeline

The system consists of a two-stage pipeline in which explainability is built into the structure rather than being added afterwards. In the first stage, which involves argument mining, the legally relevant sentences are identified from the text of the cases; in the second stage, concerned with outcome prediction, the sentences thus obtained are embedded and it is determined which ECHR articles were violated, with each prediction being traced back to specific sentences through projection onto the SVM weight vector. Although a hierarchical LegalBERT classifier is provided as an additional deep-learning method for comparison, it

is not the main contribution. The overall design of the pipeline, the two-stage approach, and the incorporation of argument mining as a pre-processing step for an interpretable classifier are the original contributions of this study.

Figure 1 gives an overview of the two-stage pipeline. The case paragraphs are split into sentences and these sentences are scored using a fine-tuned LegalBERT; those sentences which exceed the 0.90 confidence threshold are kept as premises. The premises that have been extracted are embedded using `all-mpnet-base-v2`, combined to form a 2,309-dimensional case vector, and then classified by a per-article linear SVM. When performing inference, each predicted article violation is traced back to the most contributing premises by means of projection using the SVM weight vector.

2.2 Stage 1, Argument Mining

The system is a binary sentence classifier based on `nlpaueb/bert-base-uncased-ecchr` (LegalBERT) (Chalkidis et al., 2020), a BERT model which has been pre-trained on ECtHR and EU legislation. The architecture follows the standard approach for sequence classification (using the CLS token and then a linear head); no modifications to the architecture have been made.

The training data consists of an annotation dataset containing 12,947 (case, article) pairs (azminajid, 2024), with sentence-level argument labels. The PREMISE sentences are the `arg_units` provided by the Applicant and State agents, while the NON_PREMISE sentences are the quoted Convention articles (Non-Argument units). Arguments from the ECHR agents are excluded in order to avoid outcome leakage. The dataset is divided on a case-by-case basis (70/15/15) to prevent data leakage.

The original contribution was the use of similarity-filtered factual negatives. When this method was applied directly to the LexGLUE paragraphs (which are factual summaries, not arguments put forward by the parties), the model tended to over-extract because factual sentences are more similar to party arguments than to quotations from the Convention articles — being the only NON_PREMISE examples encountered during training. In order to address this, the sentences in the facts section of the training cases were embedded using `all-mpnet-base-v2` (Reimers

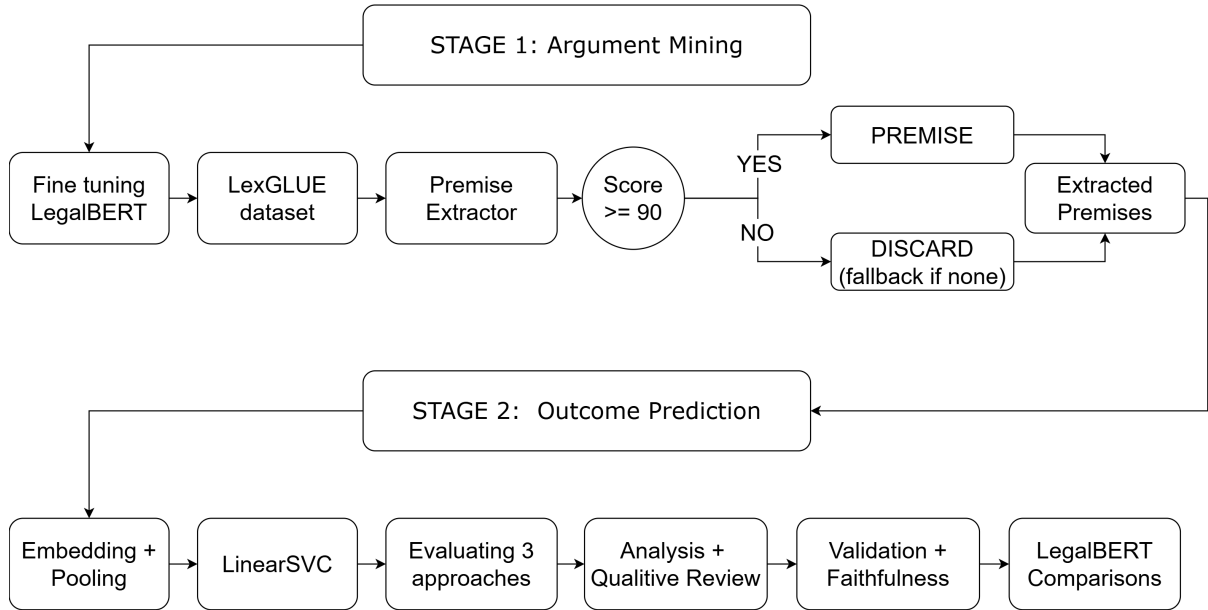


Figure 1: Overview of the two-stage pipeline. Stage 1: LegalBERT is fine-tuned on 12,947 annotated (case, article) pairs, then applied to LexGLUE paragraphs to extract premises at threshold 0.90. Stage 2: extracted premises are embedded with `all-mpnet-base-v2`, pooled into a 2,309-d vector, and classified by a per-article linear SVM; each prediction is attributed back to specific premises via weight-vector projection.

and Gurevych, 2019) and any sentence with a cosine similarity greater than 0.6 to any premise in the same case was added as a `NON_PREMISE` training example. Those facts with high similarity (which were likely of legal relevance) were discarded. The entire approach and its implementation in `fact_filter.py` are original.

A confidence threshold of 0.90 is applied to the sentence-level scores on the LexGLUE paragraphs. Figure 2 shows the score distribution is strongly bimodal: 63.8% of sentences score below 0.1, 27.0% score above 0.9, and the remaining $\approx 9.2\%$ fall in between; the threshold sits at the natural dip between the two modes and was selected from the validation distribution with no information from the test set. In instances where no premises are extracted, the entire paragraph text is used instead.

The code makes use of the HuggingFace Trainer API (Wolf et al., 2020) with the standard arguments. The training loop, the data loading, and the evaluation code found in `finetune_legalbert.py` are our own original work. The `fact_filter.py` module is also entirely original.

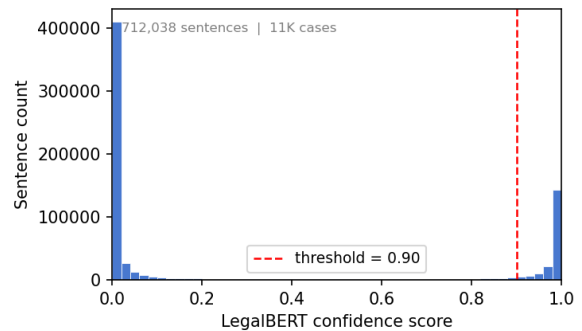


Figure 2: Distribution of LegalBERT confidence scores across all sentences in the dataset (train + val + test). The bimodal pattern — with mass concentrated below 0.10 and above 0.90 — motivates the fixed threshold of 0.90.

2.3 Stage 2, SVM Outcome Prediction

The use of embeddings involves representing each case by embedding its extracted premises using `all-mpnet-base-v2` (Reimers and Gurevych, 2019) (768 dimensions per sentence) and then applying concatenated pooling, which computes the max, the mean, and the confidence-weighted mean across all of the premise embeddings before concatenating them into a 2,304-dimensional vector. Five manually designed features are added (the fallback flag, the number of premises, the mean and maximum confidence, and the ratio

of premises to sentences), resulting in a feature vector of 2,309 dimensions. The approach to concatenated pooling and its implementation in `embedder.py` are our own.

Figure 3 shows the distribution of premises extracted per test case; the distribution is right-skewed with 49 zero-premise cases that fall back to full paragraph text.

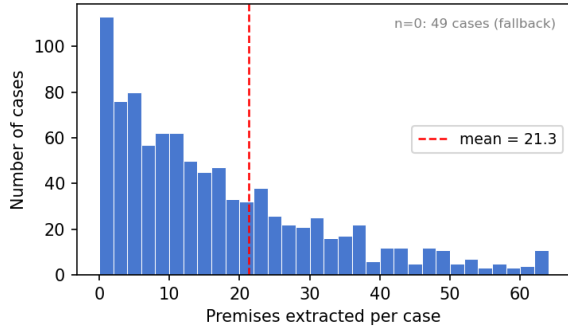


Figure 3: Number of premises extracted per test case (threshold = 0.90). The mean of 21.3 premises per case corresponds to approximately 27% of full case text. Cases with zero premises (shown in annotation) fall back to the full paragraph text.

This is a hybrid version: a second embedder, named `embedder_hybrid.py`, embeds all the paragraphs (not merely the premises) and adds to each paragraph’s vector four features that indicate awareness of the premises (`has_premise`, `n_premises`, `premise_density`, `avg_confidence`), while when pooling it gives premise-containing paragraphs a $2\times$ boost. The variant together with its implementation is original.

A `MultiOutputClassifier` that uses `LinearSVC` (Pedregosa et al., 2011) trains a separate binary classifier for each article, using balanced class weights. The per-article decision thresholds are determined on the validation set by varying the SVM decision function scores in order to maximise the per-article F_1 score. This threshold-tuning process, as carried out in `classifier.py`, is original. The LexGLUE evaluation protocol (which includes appending a ‘No Violation’ column in the case of 11-class evaluation) is based on (Chalkidis et al., 2022).

Explainability is achieved by projecting each premise embedding onto the SVM weight vector for the predicted article and ranking premises by signed dot product. Because the input rep-

resentation includes non-linear operations (max-pooling), this is an approximation rather than an exact decomposition, but it requires no separate surrogate model. The method is implemented in `traceback.py` and is an original contribution.

2.4 Stage 2, Hierarchical LegalBERT (Supplementary)

The architecture involves dividing the cases into 64 segments, each containing 128 tokens. LegalBERT encodes each segment independently in order to generate a [CLS] embedding; afterwards, a two-layer Transformer addresses the segments in a contextual manner; max pooling is then applied to obtain the case representation, which is subsequently fed into a sigmoid multi-label classifier. This hierarchical architecture has been adapted from the `hierbert.py` implementation in the LexGLUE codebase (Chalkidis et al., 2022), with some original changes made in `bert_classifier.py` relating to checkpointing, seed control, skipping over empty segments, and the persistence of the results. The three input conditions that are tested are: the full text, the premises alone, and a hybrid version consisting of the full text with [PREMISE] markers placed around the premise sentences.

Training protocol: Follows LexGLUE: $lr = 3 \times 10^{-5}$, effective batch size 8 (physical $2 \times$ gradient accumulation 4), up to 20 epochs, fp16, early-stopping patience 3. One documented deviation: checkpoints are selected on macro F_1 rather than micro F_1 as in LexGLUE. Five seeds are run per model for statistical comparisons.

3 Experimental setup and results

3.1 Datasets and Evaluation Protocol

The experiments make use of the LexGLUE `ecthr_a` benchmark (Chalkidis et al., 2022), which consists of 9,000 training cases, 1,000 validation cases and 1,000 test cases. For each case a subset of the 10 ECHR articles that have been violated is provided as an annotation. In accordance with the LexGLUE protocol, a ‘No Violation’ label is added (indicating 1 in the case where no article has been violated), so that the problem becomes an 11-class multi-label classification task. Performance is given as the macro-averaged and micro-averaged F_1 score across all 11 classes. In Stage 1 fine-tuning the separate argument annota-

tion dataset comprising 12,947 (case, article) pairs is used (azminajid, 2024); the test portion of this dataset is employed solely for reporting the accuracy of argument classification and has no role in the Stage 2 evaluation.

3.2 Stage 1, Argument Mining

Hyperparameters. The LegalBERT model is fine-tuned using the HuggingFace Trainer API with the following settings: learning rate 3×10^{-5} , batch size 16, warmup ratio 0.06, weight decay 0.01, up to 5 epochs, early stopping patience 2, fp16 on GPU. The maximum sequence length is 128 tokens. We balance training sentences by undersampling the majority class. The factual negative similarity threshold is 0.6 (cosine similarity); this value was chosen conservatively to exclude only sentences clearly dissimilar from any premise.

Results. Table 1 reports classification performance on the Stage 1 test set.

Table 1: Stage 1 argument classification results on the fine-tuning test set.

Class	P	R	F1	Support
NON_PREMISE	0.99	0.95	0.97	77,497
PREMISE	0.93	0.98	0.95	44,595
Macro avg	0.96	0.97	0.96	122,092

3.3 Stage 2, SVM Classical ML

Hyperparameters. The SentenceTransformer model used is `all-mpnet-base-v2` (it is frozen and not fine-tuned); concatenation pooling results in a 2,304-dimensional vector, to which five handcrafted features are added thereby giving a dimensionality of 2,309; the SVM makes use of `LinearSVC` with `class_weight='balanced'` and with the default regularisation (that is, $C = 1.0$); for each article, the decision threshold is varied in steps of 0.05 across the decision scores from the validation set in order to maximise the per-article F_1 score, and the best threshold for each article is then applied at test time.

Conditions.

- **Baseline:** full paragraph text, no premise extraction.

- **Premise-only:** extracted premises only (0.90 threshold).
- **Hybrid:** full paragraphs with premise-awareness features and $2 \times$ pooling boost on premise-containing paragraphs.

Results. Table 2 reports SVM performance on the LexGLUE test set.

Table 2: SVM outcome prediction results (LexGLUE `ecthr_a` test set, 1,000 cases, 11-class protocol).

Condition	Macro F1	Micro F1
Baseline (full text)	0.5669	0.6396
Premise-only	0.5193	0.5989
Hybrid	0.5680	0.6485

The full-text baseline is matched by the hybrid SVM, it also providing explanations at the sentence level. With only premises it underperforms by 0.048 macro F_1 , which shows that the cost of compression is greater than the benefit of selection when the word budget is about 27%.

3.4 Extractor Validation, Matched-Budget Controls

In order to assess whether the extractor chooses sentences that are better than those selected arbitrarily, three control conditions are examined within the same word budget (about 27% of the full text, or about 390 words per case): the complement (that is, the sentences which the extractor rejected), a random sample (sentences selected at random), and the lead (the first N sentences). The conditions are matched to the same word budget as the premises and are evaluated using the same SVM settings. For the stochastic conditions (random and complement), five different random seeds are used in order to calculate the mean and standard deviation. The results are given in Table 3.

Table 3: Matched-budget control results (SVM). Random and complement report mean $\pm$ std over 5 seeds.

Condition	Macro F1	Micro F1
Premises	0.5193	0.5989
Complement	0.5215 ± 0.015	0.5855 ± 0.017
Random	0.5008 ± 0.017	0.5828 ± 0.012
Lead	0.4713	0.5271

Premises score +1.11 standard deviations above the mean of the 5 random seeds (a modest positive

signal) but fall -0.15 standard deviations from the complement mean (no meaningful difference). With only $n = 5$ seeds these scores are indicative rather than statistically conclusive. Although the extractor is able to identify sentences that have an argument-like character, argument-likeness does not mean that there is an article-predictive signal — this is seen as a result concerning the task rather than as an indication of the extractor’s failure.

3.5 Faithfulness

The evaluation of premise attributions makes use of the ERASER framework (DeYoung et al., 2020). For each (case, predicted-article) pair, the top-three attributed premises are determined by means of SVM weight projection and are then compared with a random baseline of identical size. Two metrics are given: *comprehensiveness* (the decrease in prediction confidence when the rationale is removed) and *sufficiency* (whether the rationale on its own is able to reconstruct the prediction). The results are based on 1,180 (case, predicted-article) pairs and are presented in Table 4.

Table 4: ERASER faithfulness metrics (top-3 premises vs. random-3, 1,180 pairs).

Metric	Top-3	Random-3	Gap
Comprehensiveness ($\uparrow$)	0.2145	0.0483	+0.1662
Sufficiency ($\downarrow$)	-0.3129	0.4114	+0.7243
Prediction flips on removal ($\uparrow$)	21.4%	9.8%	+11.6%
Held by rationale alone ($\uparrow$)	86.3%	60.8%	+25.5%

Cited premises are $4.4\times$ more impactful than random ones when removed. The negative sufficiency score for the top-3 (-0.31) shows that the three cited premises alone do better than the complete set of premises for the particular predicted article, which means the other premises act as noise for that article. Since both of these measures are above their respective random baselines, this confirms that the attributions are faithful to the classifier’s decisions.

3.6 Hierarchical LegalBERT, Seed Study

Hyperparameters. In accordance with the LexGLUE protocol (Chalkidis et al., 2022), the learning rate is set at 3×10^{-5} , the effective batch size is 8 (this being equivalent to a physical batch size of 2 with a gradient accumulation of 4), training proceeds for up to 20 epochs, fp16 is used, and the early-stopping patience is 3. The architecture consists of 64 segments $\times$ 128 tokens, a 2-layer

segment Transformer, and makes use of max pooling. **Documented deviation:** the checkpoints are chosen based on macro F_1 ; LexGLUE employs micro F_1 . Five seeds are run per model; paired comparisons use the same seeds are applied to all three models.

Results. Table 5 reports per-seed and aggregate results.

Paired t -tests show hybrid vs. full-text is statistically inconclusive ($t = -0.36$, $p = 0.74$); hybrid vs. premises is significant ($t = +3.59$, $p = 0.023$, wins 5/5 seeds). No claim is made that hybrid BERT outperforms full-text BERT.

4 Discussion

4.1 Quantitative Results

Argument Mining. LegalBERT achieves a 0.96 macro F_1 (precision 0.99 for NON_PREMISE, recall 0.98 for PREMISE), effectively distinguishing arguments from quotations. However, target domain (LexGLUE) extraction quality is better assessed via the matched-budget controls.

SVM Main Results. The hybrid SVM matches the full-text baseline (0.5680 vs. 0.5669 macro F_1 , Table 2), demonstrating that explainability can be achieved without sacrificing accuracy. Conversely, the premise-only model drops by 0.048 macro F_1 , indicating that removing 73% of the text incurs a compression penalty that outweighs the selection benefits — a common finding in rationale-based models (DeYoung et al., 2020).

Extractor Controls. Premises score $+1.11$ standard deviations above the random-seed mean but only -0.15 from the complement mean. This suggests the predictive signal is distributed throughout the text, or that the extractor identifies legal argumentation rather than statistical outcome predictors. The lead baseline performs worst (0.4713), proving position alone is a poor heuristic.

Faithfulness. ERASER metrics (Table 4) confirm faithful attributions. Removing cited premises reduces prediction confidence $4.4\times$ more than random removal. Notably, top-3 sufficiency is negative (-0.31), meaning the top 3 premises alone outperform the full set of ~ 21 premises for a given article; the remaining premises act as noise.

Hierarchical LegalBERT. Full-text BERT (0.634 macro F_1 , Table 5) outperforms the SVM, prov-

Table 5: Hierarchical LegalBERT results (macro F_1 , 5 seeds, LexGLUE `ecthr_a` test set). Paired t -tests: hybrid vs. full-text $p = 0.74$ (inconclusive); hybrid vs. premises $p = 0.023$ (significant, wins 5/5 seeds).

Model	s1	s2	s3	s4	s5	Mean	Std
Full-text	0.668	0.630	0.618	0.635	0.620	0.634	0.020
Hybrid	0.643	0.603	0.623	0.654	0.632	0.631	0.020
Premises	0.586	0.579	0.619	0.596	0.575	0.591	0.017

ing end-to-end models better capture complex patterns. Hybrid and full-text BERT are statistically tied ($p = 0.74$), meaning [PREMISE] markers offer no significant advantage. Premise-only BERT is significantly weaker ($p = 0.023$), showing deep models are particularly hindered by input compression.

Per-Article Variance. Performance varies considerably by article (hybrid SVM). Articles with distinct and frequent factual patterns (Art. 3, 5, 2) score highest, while rare articles with overlapping violations (Art. 9, 14) score lowest. The ‘No Violation’ class consistently underperforms across all conditions. A full per-article breakdown is available in the notebook.

4.2 Error Analysis

No Violation. This class systematically fails ($F_1 \approx 0.22$). These cases are rare, short procedural dismissals lacking the factual narrative needed for premise extraction. They likely require a dedicated classifier or length-based features.

Rare Articles. Articles 9 and 14 have very few test cases, making their per-article F_1 highly volatile. They should be excluded from general conclusions about system performance.

Qualitative Example. The following is a correctly predicted case (TP) from the test set, showing the top-3 premises attributed to Article 10 by SVM weight-vector projection:

True: Article 10 Predicted: Article 10
 [1] (score 0.106) “Within the ambit of Article 10, freedom of speech did not protect only information and views that were...”
 [2] (score 0.076) “It was not the source, but society’s interest in free communication of news and opinions, which was...”
 [3] (score 0.054) “The purpose was not to protect the source, but rather the public interest in free communication of news and opinions...”

All three premises directly invoke Article 10 doctrine, demonstrating that the attribution identifies

legally substantive sentences rather than procedural boilerplate.

The Complement Discrepancy. Extractor-rejected sentences (the complement) are just as predictive as premises. Rejected sentences often contain highly predictive factual background (e.g., detention length for Art 5) but lack explicit argumentative markers (e.g., “the applicant submitted”). The extractor learned argument structure, not outcome relevance. Future work should use rationale supervision (DeYoung et al., 2020) derived from Stage 2 signals.

Zero-Premise Cases. In 49 cases (4.9%), no premises were extracted. These are primarily short admissibility decisions where LexGLUE provides minimal text, making predictions inherently difficult.

Attribution Quality. While overall faithful, the SVM sometimes attributes predictions to uninformative text (dates, procedural statements, or explicit mentions like “Article 6”) due to lexical correlation in the training data. Applying heuristic filters (e.g., minimum length, excluding direct article citations) before displaying rationales to users could resolve this.

5 Conclusion

The study proposed a two-stage explainable approach for predicting the legal outcomes of cases at the ECtHR. A LegalBERT model, which had been fine-tuned using similarity-filtered factual negatives, was able to extract legally relevant premises from the case text with a macro F_1 score of 0.96; an interpretable SVM was then used to predict violations of articles based on these premises, assigning each decision back to specific sentences through weight vector projection. The hybrid SVM—this being a combination of the full text with signals that take into account the premises—achieves a macro F_1 of 0.568, which is the same as that of the full-text baseline,

while also providing explanations at the sentence level. ERASER faithfulness metrics show that the premises cited are 4.4 times more influential than random ones, and 86.3% of the predictions can be reconstructed from just three of the cited sentences. A separate hierarchical LegalBERT seed study obtained a macro F_1 of 0.634 ± 0.020 , and the hybrid and full-text models were found to be statistically indistinguishable ($p = 0.74$).

The most surprising result was that the extracted premises did not perform better than their counterparts when the number of words was matched ($z = -0.15$). Although the extractor is able to reliably identify sentences that have an argument-like character, argument structure and the signal that predicts the use of articles are not identical within the LexGLUE domain — sentences which give factual background and are rejected by the extractor have just as much predictive value as the premises that are selected. This highlights a fundamental limitation: the Stage 1 model was trained to recognise the form of legal argumentation, whereas Stage 2 rewards content that correlates with the article labels. The most promising avenue for future research would be to supervise premise extraction directly on sentence-level outcome relevance rather than on argument annotation labels. Other limitations include the low performance observed for rare articles (Article 9, 14) and the No Violation class, the noted deviation from the LexGLUE checkpoint selection procedure, and the unverified source of the argument annotation dataset. Future work might also investigate the joint end-to-end training of both stages, or replace the linear SVM with a sparse attention mechanism that maintains the attributability guarantee while capturing non-linear interactions between the premises.

6 Links to external resources

- **Code repository:** https://github.com/Aghaei-Amirmasoud/explainable_by_design_ecthr_a
- **Argument annotation dataset:** <https://github.com/azminajid/ecthr-args-dataset>
- **LexGLUE benchmark:** https://huggingface.co/datasets/coastalcph/lex_glue

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